

INTRODUCTION TO COMPUTER



A **computer** is an electronic machine that manipulates data according to a list of instructions.



A **computer** is an electronic machine (device) that accepts input (data), processes it and gives (output) results (information) and saves it as an electronic file.



COMPUTER SYSTEM

Monitor + CPU + Keyboard + Mouse



Accepts Data



Processes Data



Produces Output



Stores as Electronic File

CHARACTERISTICS OF COMPUTER

A computer has several important characteristics that make it fast, accurate, reliable and useful for different types of work.



1. Speed

A computer can perform calculations and process instructions much faster than a human - millions or even billions of operations in a very short time.



2. Accuracy

A computer produces highly accurate results when the input and instructions are correct. Most errors occur due to wrong input or incorrect instructions.



3. Diligence

A computer can perform the same task repeatedly without getting tired, bored or losing concentration - maintaining the same speed and accuracy throughout.



4. Versatility

A computer can perform many different tasks - preparing documents, watching videos, browsing the internet, programming, designing and managing databases.

CHARACTERISTICS OF COMPUTER (contd.)



5. Reliability

A computer provides consistent and dependable results - given the same input and instructions under the same conditions, it will normally produce the same output every time.



6. Memory

A computer can store a large amount of data, instructions and results, and retrieve them whenever required. Computer memory is mainly divided into:

Primary Memory - RAM, ROM

Secondary Memory - Hard Disk, SSD, Pen Drive, etc.



7. Automation

Once a computer is given the required instructions and data, it can perform tasks automatically with little or no human intervention.

GENERATIONS OF COMPUTERS

From vacuum tubes to Artificial Intelligence - the evolution of computing technology across five generations



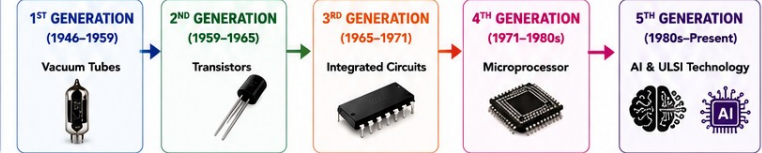
GENERATION OF COMPUTERS

Computers have been developed through different stages called generations of computers. Each generation is based on the major technology used to build computers.

DEFINITION

A generation of computer refers to a period in the development of computers, characterized by a major technological advancement in hardware, software, memory, and input/output devices which improves speed, efficiency, reliability, and ease of use.

EVOLUTION TIMELINE



1 FIRST GENERATION (1946-1959)



Examples:
ENIAC, UNIVAC I,
IBM 701, Mark I

DEFINITION

The first generation of computers used vacuum tubes as the main electronic component. These computers were very large, consumed a lot of electricity, and generated much heat.

MAIN FEATURES

- Used vacuum tube technology.
- Magnetic drums were used for memory.
- Supported machine language only.
- Very large in size.
- Required large rooms for installation.
- Consumed large amount of electricity.
- Used punch cards for input.

ADVANTAGES

- Very fast compared to earlier calculating devices.
- Could perform calculations in milliseconds.
- Made large-scale electronic computing possible.

DISADVANTAGES

- Very large in size.
- Consumed a lot of electricity.
- Generated excessive heat.
- Required air conditioning.
- Frequent maintenance.
- Vacuum tubes failed often.
- Very expensive.
- Not portable.
- Supported only machine language.
- Magnetic drums had limited storage.

2 SECOND GENERATION (1959-1965)



Examples:
IBM 1400 series,
IBM 1600 series,
UNIVAC III

DEFINITION

The second generation of computers used transistors instead of vacuum tubes. They were smaller, faster, more reliable and consumed less power.

MAIN FEATURES

- Used transistors.
- Smaller, faster and more reliable than first generation.
- Magnetic core memory used.
- Magnetic tapes and disks used for secondary storage.
- Assembly language used.
- High level languages like FORTRAN, COBOL developed.
- Punch cards used for input.
- Batch processing was used.

ADVANTAGES

- Smaller than first generation.
- More reliable.
- Consumed less electricity.
- Generated less heat.
- Faster than first generation.
- More affordable.
- Supported high level languages.
- Greater memory capacity.
- Used for commercial applications.

DISADVANTAGES

- Required cooling systems.
- Regular maintenance required.
- Manufacturing was complex.
- Used for specific applications.
- Costly.
- Punch cards still used for input.

3 THIRD GENERATION (1965-1971)



Examples:
IBM System/360,
IBM System/370,
UNIVAC 1108

DEFINITION

The third generation of computers used Integrated Circuits (ICs). ICs contain many electronic components on a single chip, making computers smaller, faster and more reliable.

MAIN FEATURES

- Used Integrated Circuits (ICs).
- Computers became smaller and more reliable.
- Semiconductor memory used.
- Magnetic disks and floppy disks used for storage.
- Keyboard used for input and monitor for output.
- More high level languages developed.
- Advanced operating systems.

ADVANTAGES

- Smaller than second generation.
- More reliable.
- Faster processing.
- Consumed less electricity.
- Generated less heat.
- Less expensive.
- Suitable for general purpose applications.
- Easier to operate.
- Commercial production easier.

DISADVANTAGES

- Manufacturing ICs required sophisticated technology.
- IC based systems were still costly.
- Maintenance required skilled technicians.

4 FOURTH GENERATION (1971-1980s)



Examples:
IBM PC, Osborne 1,
Apple II, Star 1000

DEFINITION

The fourth generation of computers used microprocessors based on LSI and VLSI technology. Personal computers became popular in this generation.

MAIN FEATURES

- Used microprocessor.
- Used LSI and VLSI technology.
- Personal computers became popular.
- Floppy disks and hard disks used.
- Keyboard and mouse used.
- Monitors used for output.
- Networking and concept of internet developed.

ADVANTAGES

- Very small in size.
- Faster processing.
- More reliable.
- More affordable.
- Consumes less electricity.
- Generates very little heat.
- Portable computers possible.
- Suitable for general purpose.
- Personal computers widely available.

DISADVANTAGES

- Advanced technology required for manufacturing.
- Skilled professionals required.
- Rapid changes made older systems obsolete.

5 FIFTH GENERATION (1980s-PRESENT)



Examples:
Laptops, Smartphones,
AI Systems, Modern PCs

DEFINITION

The fifth generation focuses on Artificial Intelligence (AI) and ULSI technology. Computers of this generation are very powerful and intelligent.

MAIN FEATURES

- Uses ULSI (Ultra Large Scale Integration) technology.
- Uses powerful microprocessors.
- Focus on Artificial Intelligence (AI).
- Supports natural language processing.
- Supports speech and image recognition.
- Uses GUI based operating systems.
- Very small and low power.
- High speed networking and internet.

ADVANTAGES

- Very high processing speed.
- Small and portable.
- Highly reliable.
- Large storage capacity.
- Low power consumption.
- Supports AI and expert systems.
- Handles large amount of data.
- User friendly GUI.
- Supports internet and global communication.

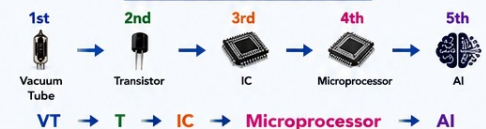
DISADVANTAGES

- Very expensive technology.
- Requires highly skilled professionals.
- Security and privacy risks.
- Dependence on computers is increasing.
- Requires large computing resources for AI.

QUICK REVISION TABLE

Generation	Period	Main Technology	Programming Languages	Examples
1st	1946-1959	Vacuum Tubes	Machine Language	ENIAC, UNIVAC
2nd	1959-1965	Transistors	Assembly, FORTRAN, COBOL	IBM 1400, UNIVAC III
3rd	1965-1971	Integrated Circuits	High Level Languages	IBM System/360, UNIVAC 1108
4th	1971-1980s	Microprocessor, LSI, VLSI	C, C++, etc.	IBM PC, Apple II
5th	1980s-Present	ULSI, AI	AI & Advanced Languages	Laptops, Smartphones, AI Systems

EASY WAY TO REMEMBER



WHAT IS A GENERATION OF COMPUTERS?

A generation refers to a period in the development of computers, characterized by a major technological advancement in hardware, memory and input/output devices - each new generation improves speed, efficiency, reliability and ease of use.

EVOLUTION TIMELINE



1

FIRST GENERATION

1946 - 1959



DEFINITION

Used vacuum tubes as the main electronic component. Very large in size, consumed a lot of electricity and generated much heat.

EXAMPLES

ENIAC, UNIVAC I, IBM 701, Mark I

MAIN FEATURES

- Vacuum tube technology
- Magnetic drums for memory
- Machine language only
- Punch cards for input

ADVANTAGES

- Very fast vs earlier devices
- Calculations in milliseconds
- Large-scale computing possible

DISADVANTAGES

- Very large, costly, huge heat
- Needed AC, frequent maintenance
- Not portable, tubes failed often

2

SECOND GENERATION

1959 - 1965



DEFINITION

Used transistors instead of vacuum tubes - smaller, faster, more reliable and consumed less power than 1st generation.

EXAMPLES

IBM 1400 series, IBM 1600 series, UNIVAC III

MAIN FEATURES

- Transistors, magnetic core memory
- Magnetic tapes/disks for storage
- Assembly language used
- FORTRAN, COBOL developed

ADVANTAGES

- Smaller, more reliable, faster
- Consumed less electricity
- Supported high level languages

DISADVANTAGES

- Needed cooling systems
- Complex manufacturing, costly
- Punch cards still used for input



THIRD GENERATION

1965 - 1971



DEFINITION

Used Integrated Circuits (ICs) that pack many electronic components on a single chip, making computers smaller and faster.

EXAMPLES

IBM System/360, IBM System/370, UNIVAC 1108

MAIN FEATURES

- Integrated Circuits (ICs)
- Semiconductor memory
- Keyboard input, monitor output
- Advanced operating systems

ADVANTAGES

- Smaller, more reliable
- Faster, less electricity used
- Easier to operate commercially

DISADVANTAGES

- Sophisticated manufacturing needed
- IC based systems still costly
- Needed skilled technicians



FOURTH GENERATION

1971 - 1980s



DEFINITION

Used microprocessors based on LSI and VLSI technology. Personal computers became popular in this generation.

EXAMPLES

IBM PC, Osborne 1, Apple II, Star 1000

MAIN FEATURES

- Microprocessor, LSI/VLSI tech
- Floppy & hard disks used
- Keyboard and mouse
- Networking, internet concept

ADVANTAGES

- Very small, faster, affordable
- Consumes very little power
- Portable computers possible

DISADVANTAGES

- Advanced manufacturing tech needed
- Skilled professionals required
- Rapid change made systems obsolete

5

FIFTH GENERATION

1980s - Present



DEFINITION

Focuses on Artificial Intelligence (AI) and ULSI technology. Computers of this generation are very powerful and intelligent.

EXAMPLES

Laptops, Smartphones, AI Systems, Modern PCs

MAIN FEATURES

- ULSI technology
- Natural language processing
- Speech & image recognition
- GUI based operating systems

ADVANTAGES

- Very high processing speed
- Small, portable, highly reliable
- Supports AI & global communication

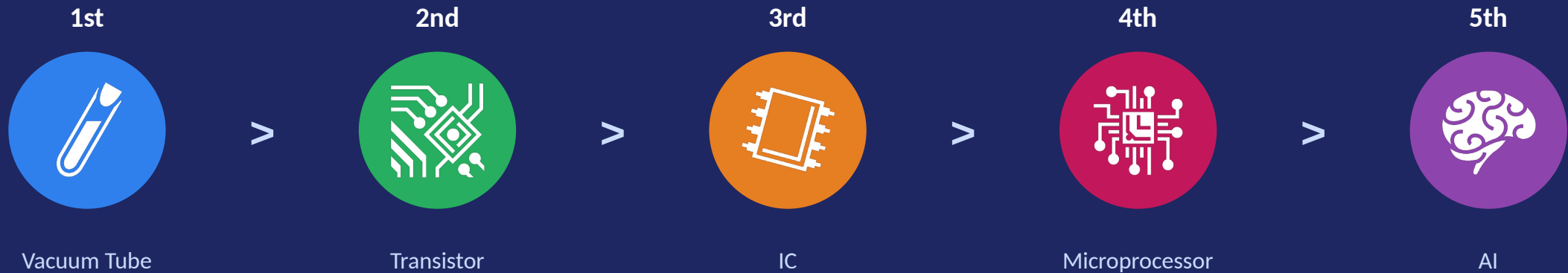
DISADVANTAGES

- Very expensive technology
- Needs highly skilled professionals
- Security & privacy risks

QUICK REVISION TABLE

Gen	Period	Main Tech	Languages	Examples
1st	1946-1959	Vacuum Tubes	Machine Language	ENIAC, UNIVAC
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3rd	1965-1971	Integrated Circuits	High Level Languages	IBM System/360, UNIVAC 1108
4th	1971-1980s	Microprocessor, LSI/VLSI	C, C++, etc.	IBM PC, Apple II
5th	1980s-Present	ULSI, AI	AI / Advanced Languages	Laptops, Smartphones

EASY WAY TO REMEMBER



VT > T > IC > Microprocessor > AI

Tip: Remember the trend - each generation got smaller, faster, cheaper and more powerful than the one before it.

BLOCK DIAGRAM OF COMPUTER

1 INPUT UNIT

Definition:

The Input Unit consists of input devices that allow you to enter data into your computer.

Examples:

Keyboard, Mouse, Joystick, Scanner, etc.



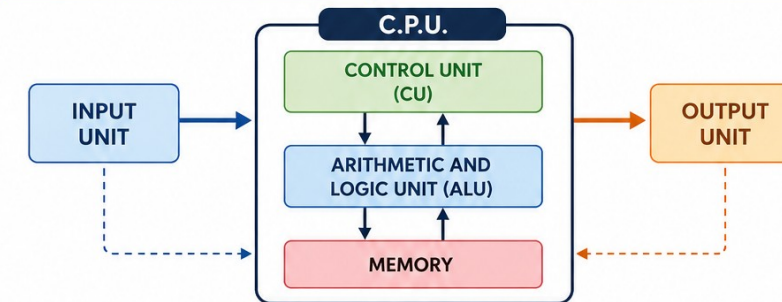
2 CENTRAL PROCESSING UNIT (CPU)

Definition:

The CPU is known as the **brain of the computer system**. Once the information is entered into the computer by the input device, the CPU processes it.

It consists of three components:

- Arithmetic and Logic Unit (ALU)
- Control Unit (CU)
- Memory



3 ARITHMETIC AND LOGIC UNIT (ALU)

Definition:

ALU performs arithmetic and logical operations, such as addition, subtraction, multiplication, division, etc.



4 CONTROL UNIT (CU)

Definition:

As the name suggests, the Control Unit controls all the activities and operations of the computer system.



5 MEMORY

Definition:

Memory is a storage media which is used to store data and instructions before and after processing.

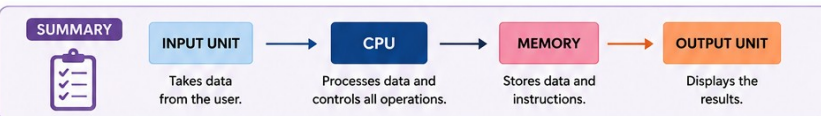


6 OUTPUT UNIT

Definition:

The Output Unit consists of output devices that are used to display the results or output.

Examples: Monitor, Printer.



TYPES OF COMPUTERS AND THEIR FEATURES

Computers can be classified into different types based on their size, processing speed, storage capacity, performance and number of users.

1 MINI COMPUTERS

Definition :

A Mini Computer is a medium-sized computer that is more powerful than a microcomputer but smaller and less powerful than a mainframe computer.

Features :

- Medium processing speed.
- Supports multiple users at the same time.
- More powerful than microcomputers.
- Has greater storage capacity than microcomputers.
- Used in small and medium-sized organizations.
- Can perform multiple tasks simultaneously.

Examples : PDP-11, IBM AS/400.

Exam Point : Mini Computer → Medium-sized, multi-user and more powerful than a microcomputer.



2 MICRO COMPUTERS

Definition :

A Micro Computer is a small computer that uses a microprocessor as its CPU and is generally designed for use by one person at a time.

Features :

- Small in size.
- Low cost compared with larger computers.
- Designed mainly for a single user.
- Easy to operate.
- Requires less space.
- Suitable for personal and office use.

Examples : Desktop Computer, Laptop, Tablet, Smartphone.

Exam Point : Micro Computer → Small, low-cost and generally single-user computer.



3 MAINFRAME COMPUTERS

Definition :

A Mainframe Computer is a large and powerful computer designed to process huge amounts of data and support many users simultaneously.

Features :

- Very high processing capacity.
- Supports hundreds or thousands of users.
- Large storage capacity.
- Highly reliable and secure.
- Can process large volumes of data.
- Used by banks, government organizations, airlines, and large companies.

Examples : IBM zSeries mainframes.

Exam Point : Mainframe Computer → Powerful multi-user computer used for large-scale data processing.



4 SUPER COMPUTERS

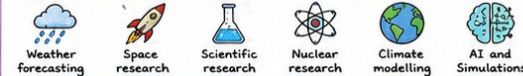
Definition :

A Super Computer is the fastest and most powerful type of computer, designed to perform extremely complex calculations and process huge amounts of data at very high speed.

Features :

- Extremely high processing speed.
- Very large storage capacity.
- Performs complex calculations very quickly.
- Uses parallel processing.
- Very expensive to build and maintain.
- Used for scientific and engineering applications.

Applications :



Exam Point : Super Computer → Extremely powerful and fastest computer used for complex calculations.



COMPARISON TABLE

Type	Size	Users	Processing Speed	Storage Capacity	Main Feature	Common Use
Micro Computer	Small	Usually one	Low to Medium	Small	Personal use	Home, Office
Mini Computer	Medium	Multiple	Medium	Medium	Medium-scale processing	Organizations
Mainframe	Large	Many	High	Large	Huge data processing	Banks, Airlines, Large Companies
Super Computer	Very Large / High-performance	Specialized	Very High	Very Large	Extremely fast calculations	Research, Weather, Scientific Work

Remember

- Micro → Personal
- Mini → Medium / Multiple Users
- Mainframe → Massive Data / Many Users
- Super → Scientific / Highest Speed

≡ TYPES OF PROGRAMMING LANGUAGES ≡

★ (Machine Languages, Assembly Languages, High Level Languages)

LANGUAGE

- Any medium to communicate is called a language.

PROGRAMMING LANGUAGE

- Any language that is used to communicate with a computer or a machine is called Programming language. Programming Language is in the form of a set of instructions.

TYPES OF PROGRAMMING LANGUAGE

- 1 Low Level / Binary language [0,1]
- 2 Assembly / Mid / Bit code language
- 3 High Level language



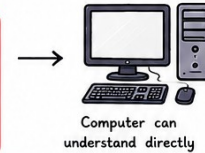
1. LOW LEVEL LANGUAGE

- Low level Programming language is the first programming language, which was used to instruct computers. It is in binary form [0,1], so it is called Binary Language. A computer can directly understand low level language but it's difficult for a human to understand.

Example :

Code

```
10110010 11001100 10101010
11001010 10110011 11001101
10101010 11001010 10110011
11001100 10101010 11001010
```



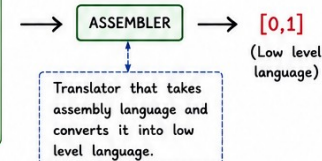
2. ASSEMBLY / MID / BIT CODE LANGUAGE

- Mid level language is called Assembly level language. In Assembly level language, predefined words are used to make programming easy. In Assembly level programs, work was easier than low level language, but computers can't understand assembly level language directly, and also not humans.
- A computer can't understand assembly level language directly, so an Assembler is used in between as a translator, which takes assembly language and converts it into low level language.

Example :

Code

```
MOV A, B
ADD A, C
SUB A, D
MUL A, E
DIV A, F
```



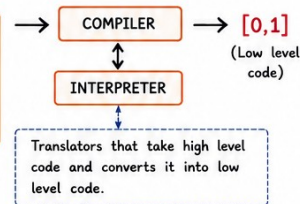
3. HIGH LEVEL LANGUAGE

- A language that a human can understand directly is called high level language. In high level language basic English is used, which makes programming very easy to understand.
- 1972 → C language
- A computer can't understand high level language code directly, so a Compiler and Interpreter are used in between as a translator, which takes high level code and converts it into low level code.
- Every high level language will have its own compiler.
- Example: JavaC is the compiler of Java.

Example :

Code

```
int a = 10;
int b = 20;
int sum = a + b;
print(sum);
```



COMPARISON TABLE

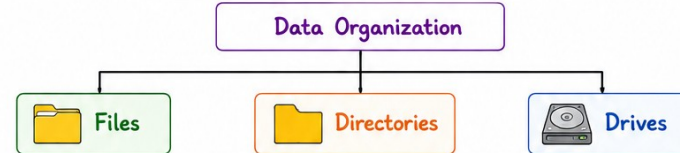
Type	Form	Understand by	Translator Used	Example
Low Level / Binary	Binary [0,1]	Computer	No Translator (Directly)	Machine Code
Assembly / Mid / Bit Code	Predefined words (Mnemonics)	Neither Computer nor Human Easily	Assembler	MOV, ADD, SUB
High Level	Basic English (Easy Words)	Human	Compiler / Interpreter	C, Java, Python

DATA ORGANIZATION

or Data Organization, Drives, Files, Directories.

It is the way of organizing data in the most efficient manner, so that it can be easily accessed whenever required. It is the arrangement of data in the most possible orderly and logical manner. Data can be arranged into sheets with headings or into graphs, charts, or tables.

Data Organization is divided into three parts:



1 Files



- Files are places where we store our data with identifiers and names.
- Data can be of any type: audio, video, text, image etc.
- Every file has its own extension, for example .jpg for image, .mov for videos, .txt for text etc.
- Two files with the same name can't exist.
- Some common functions performed on a file are creation, modification, deletion, updation etc.

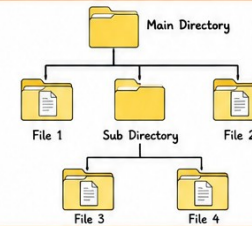
Examples of File Extensions

- .jpg → Image file
- .mov → Video file
- .txt → Text file
- .mp3 → Audio file

2 Directories



- Directories are also known as folders; they are like containers that contain all the related files.
- It is a collection of files.
- A directory can have a directory inside it, which is called a sub directory.
- It stores all the related files, so that the data can be easily fetched whenever needed.



3 Drives



- Drives are also known as disk drives.
- Drives are the storage location in your computer where data and information are kept.
- Whenever you save any data, it gets stored in one of those drive locations, so you can use it later.

Example:
Floppy drive, Primary drive, Drive D and F etc.

Common Drives in Computer



QUICK REVISION

Term	Meaning
Data Organization	Arranging data in the most efficient, orderly and logical manner.
File	A place where data is stored with a name and extension.
Directory	A folder that contains and organizes related files.
Drive	A storage location in a computer where data and information are kept.

Easy To Remember
Drive → Directory → File
(Drive contains Directories,
and Directories contain Files.)

TYPES OF MEMORY

(Primary And Secondary) RAM, ROM, PROM, EPROM,
Secondary Storage Devices (FD, CD, HD, Pen drive)
I/O Devices (Scanners, Plotters, LCD, Plasma Display)

1. MEMORY

It is a storage device which is used to store data for further use. It can be broadly classified into Primary and Secondary memory.

3. RAM (Random Access Memory)

RAM is a temporary and volatile memory that stores data and instructions currently being used by the computer.



- Temporary storage.
- Volatile memory.
- Data is lost when power is turned off.
- Faster than secondary storage.
- Used during normal computer operation.

Exam Point: RAM → Temporary + Volatile + Fast

4. ROM (Read Only Memory)

ROM is a permanent and non-volatile memory that stores important instructions required for starting and operating the computer.



- Permanent storage.
- Non-volatile memory.
- Data remains stored even when power is turned off.
- Used for the startup/booting process of a computer.
- Writing data is slower compared with RAM.

5. PROM (Programmable ROM)

PROM is a type of ROM that can be programmed only once by the user. Once programmed, its data cannot normally be changed or erased.

Exam Point: PROM → Programmable only once.

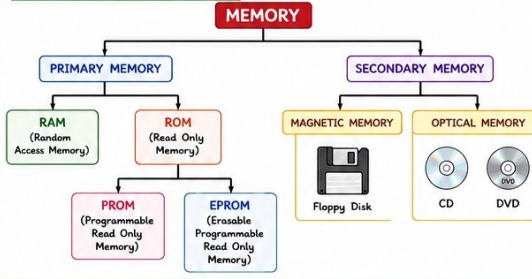
6. EPROM (Erasable Programmable ROM)



EPROM is a type of ROM whose stored can be erased and reprogrammed using ultraviolet (UV) light.

Exam Point: EPROM → Erasable + Reprogrammable using UV light.

2. MEMORY CLASSIFICATION



7. SECONDARY MEMORY

Secondary Memory is a type of non-volatile memory used to store large amounts of data and information permanently for future use.



- Also known as Auxiliary Memory.
- Does not directly communicate with the CPU.
- Slower than primary memory.
- Has a larger storage capacity.
- Non-volatile.
- Data remains stored even when the computer is switched off.

Exam Point: Secondary Memory → Large & Permanent Storage

8. SECONDARY STORAGE DEVICES

A. FLOPPY DISK (FD)

- A Floppy Disk is a magnetic storage device used to store and transfer data.
- Portable storage device.
- Uses magnetic technology.
- Has very low storage capacity compared with modern storage devices.

B. CD (COMPACT DISC)

- A CD is an optical storage device used to store and read digital data using laser technology.
- Optical storage device.
- Uses laser technology.
- Used for storing software, music, videos, and other data.

C. HARD DISK (HD)

- A Hard Disk is a secondary storage device used to store large amounts of data, programs, and files permanently.
- Large storage capacity.
- Non-volatile.
- Used for permanent data storage.

D. PEN DRIVE

- A Pen Drive is a small, portable flash-memory storage device used to store and transfer data from one computer to another.
- Portable.
- Easy to use.
- Uses flash memory.
- Data can be easily transferred.

9. I/O DEVICES (Input / Output Devices)

I/O devices are used to enter data into a computer or receive information from a computer.

A. SCANNER (Input Device)

A Scanner is an input device that captures text, images, or documents and converts them into digital form.

B. PLOTTER (Output Device)

A Plotter is an output device used to produce high-quality drawings, graphs, maps, and large technical designs on paper.

C. LCD (Output Device)

LCD (Liquid Crystal Display) is a flat-panel display technology used to display text, images, and video on a computer or other electronic device.

D. PLASMA DISPLAY (Output Device)

A Plasma Display is a flat-panel display that uses small cells containing electrically charged gases to produce images and videos.

10. RAM vs ROM

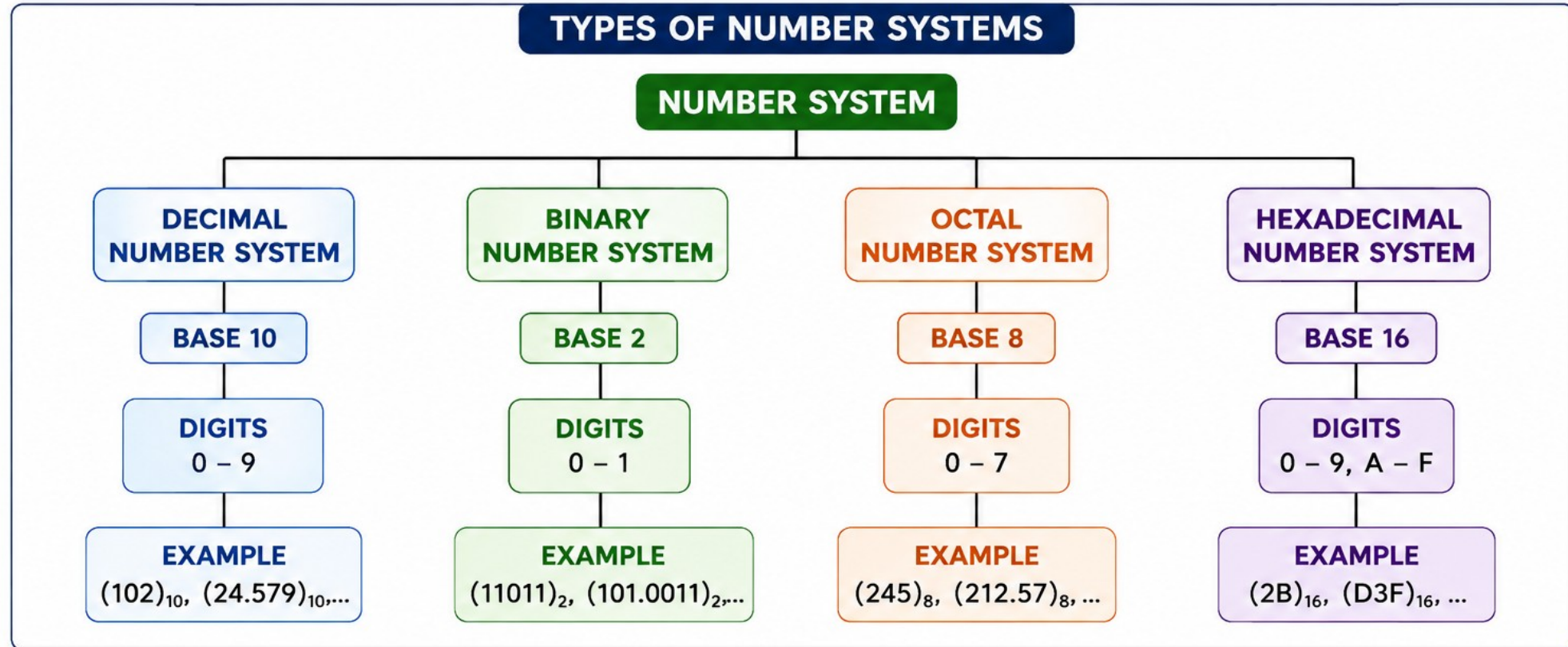
RAM	ROM
Temporary storage	Permanent storage
Store data in MBs	Store data in GBs
Volatile	Non-volatile
Used in normal operation	Used for startup process of computer
Writing data is faster	Writing data is slower

11. PRIMARY MEMORY vs SECONDARY MEMORY

PRIMARY MEMORY		SECONDARY MEMORY
It is also known as main memory	Name	It is also known as auxiliary memory
Directly communicates with the CPU	CPU Communication	It doesn't directly communicate with CPU
Faster in speed	Speed	Slower in speed
Smaller in size, which means can store limited amount of data	Size & Capacity	Larger in size, which means can hold large capacity of data
Volatile memory, which means store data until the computer is on	Volatility	Non-volatile memory, which means can store data even if the computer is off
Temporary memory	Nature	Permanent memory

≡ INTRODUCTION TO NUMBER SYSTEM ≡

Number system is a method of representing numbers with the help of a set of symbols and rules; it is a mathematical notation used to represent quantities or values in various forms.



IMPORTANT POINTS

- ▶ Humans understand decimal, and computers understand binary, so we have two different number systems.
- ▶ Binary can be long for a large amount of data, so there are more systems like octal and hexadecimal that allow binary sequences to be broken into groups, making them less error-prone and more readable compared to binary.

NUMBER SYSTEMS

Introduction to Binary, Octal, Hexadecimal System

1 DECIMAL NUMBER SYSTEM

The Decimal Number System is the standard system for denoting numbers.

- It is also called the base-10 system.
- Digits used in it are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.
- Each digit in the number is at a specific place value that is powers of 10.
- From right to left – units has the place value as 10^0 , Tens has the place value as 10^1 , Hundreds as 10^2 , Thousands as 10^3 , and so on.

PLACE VALUE (Base 10)

Thousands	Hundreds	Tens	Units
10^3	10^2	10^1	10^0
1	0	2	8

Example :

10285 can be written as
 $10285 = (1 \times 10^4) + (0 \times 10^3) + (2 \times 10^2) + (8 \times 10^1) + (5 \times 10^0)$
 $10285 = 1 \times 10000 + 0 \times 1000 + 2 \times 100 + 8 \times 10 + 5 \times 1$
 $10285 = 10000 + 0 + 200 + 80 + 5$
 $10285 = 10285$

2 BINARY NUMBER SYSTEM

Binary Number System is the number system with base 2.

- The numbers are formed using two digits – 0 and 1.
- Binary number system is very useful in electronic devices and computer systems because it can be easily performed using just two state i.e. 0 and 1.
- Each digit in the number is at a specific place value that is powers of 2.
- From right to left – as powers of 2 i.e. 2^0 , 2^1 , 2^2 , etc.

💡 Binary Numbers can be converted to Decimal value by multiplying each digit with the place value and then adding the result.

PLACE VALUE (Base 2)

2^3	2^2	2^1	2^0
1	0	1	1

Example :

$(1011)_2$ can be written as
 $(1011)_2 = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$
 $(1011)_2 = 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1$
 $(1011)_2 = 11$ (in decimal)

3 OCTAL NUMBER SYSTEM

Octal Number System is the number system with base 8.

- The numbers are formed using 8 digits i.e. 0, 1, 2, 3, 4, 5, 6, 7.
- Octal number system is useful for representing file permissions in Unix/Linux operating systems.
- Each digit in the number is at a specific place value that is powers of 8.
- From right to left – as powers of 8 i.e. 8^0 , 8^1 , 8^2 , etc.

💡 Octal Numbers can be converted to Decimal value by multiplying each digit with the place value and then adding the result.

PLACE VALUE (Base 8)

8^2	8^1	8^0
3	2	5

Example :

$(325)_8$ can be written as
 $(325)_8 = 3 \times 8^2 + 2 \times 8^1 + 5 \times 8^0$
 $(325)_8 = 192 + 16 + 5$
 $(325)_8 = 213$ (in decimal)

4 HEXADECIMAL NUMBER SYSTEM

Hexadecimal Number System is the number system with base 16.

- The numbers are formed using 16 digits i.e. 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E and F.
- Hexadecimal Numbers are useful for handling memory address locations.
- The digits from 0 to 9 are used as in the decimal system, but the numbers 10 to 15 are represented using the letters A to F as follows:

10 = A, 11 = B, 12 = C, 13 = D, 14 = E, 15 = F

💡 Hexadecimal Number System can be converted to Decimal value by multiplying each digit with the place value and then adding the result.

PLACE VALUE (Base 16)

16^1	16^0
2	F

Example :

$(2F)_{16}$ can be written as
 $(2F)_{16} = 2 \times 16^1 + F \times 16^0$
 $(2F)_{16} = 2 \times 16 + 15 \times 1$
 $(2F)_{16} = 32 + 15$
 $(2F)_{16} = 47$ (in decimal)

NUMBER SYSTEM	BASE (RADIX)	DIGITS USED	PLACE VALUE (FROM RIGHT TO LEFT)	EXAMPLE	USES
Decimal	10	0 – 9	10^0 , 10^1 , 10^2 , 10^3 , ...	$(10285)_{10}$	General counting in everyday life
Binary	2	0 – 1	2^0 , 2^1 , 2^2 , 2^3 , ...	$(1011)_2$	Used in computers and digital electronics
Octal	8	0 – 7	8^0 , 8^1 , 8^2 , 8^3 , ...	$(325)_8$	Used in UNIX/Linux file permissions
Hexadecimal	16	0 – 9, A – F	16^0 , 16^1 , 16^2 , 16^3 , ...	$(2F)_{16}$	Used in memory addresses, color codes, etc.

NUMBER SYSTEM CONVERSIONS

1 Decimal → Binary

2 Binary → Decimal

3 Decimal → Octal

4 Octal → Decimal

5 Octal → Binary

6 Binary → Octal

1 DECIMAL TO BINARY CONVERSION

METHOD

- Divide the decimal number by 2.
- Write the remainder.
- Continue until quotient becomes 0.
- Read remainders from bottom to top.

Example : $198_{10} = (?)_2$

$198 \div 2 = 99$ R 0
 $99 \div 2 = 49$ R 1
 $49 \div 2 = 24$ R 1
 $24 \div 2 = 12$ R 0
 $12 \div 2 = 6$ R 0
 $6 \div 2 = 3$ R 0
 $3 \div 2 = 1$ R 1
 $1 \div 2 = 0$ R 1

Read Bottom to Top

$198_{10} = 11000110_2$

DECIMAL TO BINARY – 10 EXAMPLES

No.	Decimal	Binary	No.	Decimal	Binary
1.	198	11000110 ₂	6.	625	1001110001 ₂
2.	13	1101 ₂	7.	53.625	110101.101 ₂
3.	231	11100111 ₂	8.	516.16	1000000100.00101000111 ₂
4.	932	1110100100 ₂	9.	156.25	10011100.01 ₂
5.	85	1010101 ₂	10.	96	1100000 ₂

Fractional Part Conversion
(Multiply by 2)

$0.625 \times 2 = 1.25 \rightarrow 1$
 $0.25 \times 2 = 0.50 \rightarrow 0$
 $0.50 \times 2 = 1.00 \rightarrow 1$

Therefore,

$53.625_{10} = 110101.101_2$

2 BINARY TO DECIMAL CONVERSION

METHOD

- Multiply every binary digit by its corresponding power of 2 (from right to left starting with 2⁰).
- Add all the results.

Example : $(10001)_2 = (?)_{10}$

$= 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$
 $= 16 + 0 + 0 + 0 + 1$
 $= 17$

$(10001)_2 = (17)_{10}$

BINARY TO DECIMAL – 7 EXAMPLES

No.	Binary	Decimal	No.	Binary	Decimal
1.	10001 ₂	17 ₁₀	5.	100110 ₂	38 ₁₀
2.	101010 ₂	42 ₁₀	6.	11000110 ₂	198 ₁₀
3.	10101011 ₂	171 ₁₀	7.	1101001101 ₂	845 ₁₀
4.	1101 ₂	13 ₁₀			

Important Powers of 2

2 ⁰	2 ¹	2 ²	2 ³	2 ⁴	2 ⁵	2 ⁶	2 ⁷	2 ⁸	2 ⁹
1	2	4	8	16	32	64	128	256	512

3 DECIMAL TO OCTAL CONVERSION

METHOD

- Divide the decimal number by 8.
- Write the remainder.
- Continue until quotient becomes 0.
- Read remainders from bottom to top.

Example : $156_{10} = (?)_8$

$156 \div 8 = 19$ R 4
 $19 \div 8 = 2$ R 3
 $2 \div 8 = 0$ R 2

Read Bottom to Top

$156_{10} = 234_8$

DECIMAL TO OCTAL – 7 EXAMPLES

No.	Decimal	Octal	No.	Decimal	Octal
1.	156	234 ₈	5.	125	175 ₈
2.	25	31 ₈	6.	198	306 ₈
3.	64	100 ₈	7.	255	377 ₈
4.	100	144 ₈			

Important Powers of 8

8 ⁰	8 ¹	8 ²	8 ³	8 ⁴	8 ⁵	8 ⁶
1	8	64	512	4096	32768	262144

4 OCTAL TO DECIMAL CONVERSION

METHOD

- Multiply each octal digit by its corresponding power of 8 (from right to left starting with 8⁰).
- Add all the results.

Example : $(234)_8 = (?)_{10}$

$= 2 \times 8^2 + 3 \times 8^1 + 4 \times 8^0$
 $= 2 \times 64 + 3 \times 8 + 4 \times 1$
 $= 128 + 24 + 4$
 $= 156$

$(234)_8 = (156)_{10}$

OCTAL TO DECIMAL – 7 EXAMPLES

No.	Octal	Decimal	No.	Octal	Decimal
1.	234 ₈	156 ₁₀	5.	175 ₈	125 ₁₀
2.	31 ₈	25 ₁₀	6.	306 ₈	198 ₁₀
3.	100 ₈	64 ₁₀	7.	377 ₈	255 ₁₀
4.	144 ₈	100 ₁₀			

5 OCTAL TO BINARY CONVERSION

METHOD

- Convert each octal digit into its equivalent 3-bit binary number.

OCTAL TO BINARY TABLE

Octal	Binary (3 bits)	Octal	Binary (3 bits)
0	000	4	100
1	001	5	101
2	010	6	110
3	011	7	111

OCTAL TO BINARY – 7 EXAMPLES

No.	Octal	Binary	No.	Octal	Binary
1.	234 ₈	10011100 ₂	5.	175 ₈	1111101 ₂
2.	31 ₈	11001 ₂	6.	306 ₈	11000110 ₂
3.	100 ₈	1000000 ₂	7.	377 ₈	11111111 ₂
4.	144 ₈	1100100 ₂			

6 BINARY TO OCTAL CONVERSION

METHOD

- Starting from the right side, divide binary digits into groups of 3 bits.
- If the leftmost group has less than 3 bits, pad zeros on the left.
- Convert each 3-bit group to its octal equivalent.

Example : $(10011100)_2 = (?)_8$

Step 1 : Group into 3-bits from right

10 011 100

Step 2 : Add leading zero to left group

010 011 100

Step 3 : Convert each group to octal

2 3 4

$(10011100)_2 = (234)_8$

BINARY TO OCTAL – 7 EXAMPLES

No.	Binary	Octal	No.	Binary	Octal
1.	10011100 ₂	234 ₈	5.	1111101 ₂	175 ₈
2.	11001 ₂	31 ₈	6.	11000110 ₂	306 ₈
3.	1000000 ₂	100 ₈	7.	11111111 ₂	377 ₈
4.	1100100 ₂	144 ₈			



IMPORTANT NOTES

- 1 octal digit = 3 binary bits (000 to 111)
- 1 binary digit = power of 2
- 1 octal digit = power of 8

ALL CONVERSION RULES – QUICK REFERENCE

DECIMAL → BINARY

$\div 2$

Divide by 2 and write remainder. Read remainder from bottom to top.

BINARY → DECIMAL

$\times 2^n$

Multiply each bit by power of 2 and add all results.

DECIMAL → OCTAL

$\div 8$

Divide by 8 and write remainder. Read remainder from bottom to top.

OCTAL → DECIMAL

$\times 8^n$

Multiply each digit by power of 8 and add all results.

OCTAL → BINARY

(3 bits)

Convert each octal digit into its equivalent 3-bit binary number.

BINARY → OCTAL

(3 bits)

Make groups of 3 bits from right and convert each group to octal.

OCTAL ↔ BINARY SHORT TRICK

Octal to Binary → Replace each octal digit by 3-bit binary.
Binary to Octal → Make 3-bit groups from right and convert.

BINARY ARITHMETIC OR CONVERSION

Simple Addition, Subtraction, Multiplication

1 ADDITION

Binary addition is performed in the same manner as decimal addition. However, since binary number system has only two digits, the addition table for binary arithmetic is very simple, consisting of only four entries.

The complete table for binary addition is:

A	B	SUM	CARRY
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Rules

- $0 + 0 = 0$
- $0 + 1 = 1$
- $1 + 0 = 1$
- $1 + 1 = 0$, plus a carry of 1 to next higher column.

EXAMPLES

1) $(1011)_2 + (1101)_2$

$$\begin{array}{r} 1011 \\ + 1101 \\ \hline 11000 \end{array}$$

$(1011)_2 + (1101)_2 = (11000)_2$

2) $(11101)_2 + (10111)_2$

$$\begin{array}{r} 11101 \\ + 10111 \\ \hline 110100 \end{array}$$

$(11101)_2 + (10111)_2 = (110100)_2$

3) $(100111)_2 + (11100)_2$

$$\begin{array}{r} 100111 \\ + 11100 \\ \hline 1001011 \end{array}$$

$(100111)_2 + (11100)_2 = (1001011)_2$

2 SUBTRACTION

Binary subtraction is the process of finding the difference between two binary numbers using the base-2 number system. It follows rules similar to decimal subtraction but limited to digits 0 and 1. It borrows 1 from the next higher order bit to the left.

A	B	A - B	BORROW
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Rules

- $1 - 0 = 1$
- $1 - 1 = 0$
- $0 - 0 = 0$
- $0 - 1 = 1$, with a borrow

EXAMPLES

1) $(10110)_2 - (01101)_2$

$$\begin{array}{r} 10110 \\ - 01101 \\ \hline 01001 \end{array}$$

$(10110)_2 - (01101)_2 = (01001)_2$

2) $(11001)_2 - (1011)_2$

$$\begin{array}{r} 11001 \\ - 1011 \\ \hline 10010 \end{array}$$

$(11001)_2 - (1011)_2 = (10010)_2$

3) $(100000)_2 - (1111)_2$

$$\begin{array}{r} 100000 \\ - 1111 \\ \hline 011001 \end{array}$$

$(100000)_2 - (1111)_2 = (011001)_2$

3 MULTIPLICATION

Binary multiplication is the process of finding the product of two binary numbers using the base-2 number system. It follows the same principle as decimal multiplication but uses only the digits 0 and 1.

Basic Rules of Binary Multiplication

A	B	A × B
0	0	0
0	1	0
1	0	0
1	1	1

Basic Products

- $0 \times 0 = 0$
- $0 \times 1 = 0$
- $1 \times 0 = 0$
- $1 \times 1 = 1$

EXAMPLES

1) $11_2 \times 11_2$ ($3 \times 3 = 9$ in decimal)

$$\begin{array}{r} 11 \\ \times 11 \\ \hline 11 \quad (11 \times 1) \\ 11 \quad (11 \times 1 \text{ shifted left}) \\ \hline 1001 \end{array}$$

$11_2 \times 11_2 = 1001_2$

2) $1011_2 \times 101_2$ ($11 \times 5 = 55$ in decimal)

$$\begin{array}{r} 1011 \\ \times 101 \\ \hline 1011 \quad (1011 \times 1) \\ 0000 \quad (1011 \times 0 \text{ shifted left}) \\ 1011 \quad (1011 \times 1 \text{ shifted left twice}) \\ \hline 110111 \end{array}$$

$1011_2 \times 101_2 = 110111_2$

3) $101_2 \times 101_2$ ($5 \times 5 = 25$ in decimal)

$$\begin{array}{r} 101 \\ \times 101 \\ \hline 101 \quad (101 \times 1) \\ 000 \quad (101 \times 0 \text{ shifted left}) \\ 101 \quad (101 \times 1 \text{ shifted left twice}) \\ \hline 11001 \end{array}$$

$101_2 \times 101_2 = 11001_2$